PREP 1 · LESSONS 1–2

Inside the atom

Protons, neutrons and electrons · atomic and mass numbers · isotopes and ions

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons · 1 mark each

1 Define electric current. DN 90 [1]

The rate of flow of electrical charge.

2 Name the two quantities that kinetic energy depends on. DN 643 [1]

Mass and speed

3 State the frequency and potential difference of the UK mains electricity supply. DN 132 [1]

50 Hz and about 230 V

Key ideas LESSONS 1–2

Show your working

4 Complete the table. Choose from: **+1 0 −1 1 very small**

Particle	Relative charge	Relative mass
Proton	+1	1
Neutron	0	1
Electron	−1	very small

Fill in the three gaps in the table. All three ✓✓; two ✓ [2]

5 An atom of sodium is $^{23}_{11}Na$

Number of protons: 11 neutrons: 12 electrons: 11 All three ✓✓; two ✓ [2]

6 Define isotopes. [1]

Atoms of the same element with different numbers of neutrons ✓

7 Which nucleus is an isotope of $^{12}_{6}C$? $^{14}_{7}N$ | (**$^{14}_{6}C$**) | $^{12}_{5}B$ [1]

8 When an atom loses an outer electron, it becomes a (**positive**) | **negative** ion. [1]